

Important MySQL Features & Advanced Constraints

1. AUTO_INCREMENT (Automatic Numbering)

When you have a Primary Key like **StudentID** or **OrderID**, it is difficult to type the numbers (1, 2, 3...) manually every time you insert a new row. MySQL can do this automatically for you using **AUTO_INCREMENT**.

- **Rule:** It is mostly used with Integer (**INT**) Primary Keys.
- **How to use it:**

```
CREATE TABLE Users (  
    UserID INT AUTO_INCREMENT PRIMARY KEY,  
    UserName VARCHAR(50) NOT NULL  
);  
  
-- When inserting data, you DO NOT need to insert the UserID.  
-- MySQL will automatically make Kamal = 1 and Nimal = 2.  
INSERT INTO Users (UserName) VALUES ('Kamal');  
INSERT INTO Users (UserName) VALUES ('Nimal');
```

2. Composite Primary Keys (Two Columns = One Key)

Sometimes, a single column is not enough to uniquely identify a row. In such cases, we can combine **two or more columns** to create a single Primary Key. This is called a Composite Primary Key. It is very common in junction tables (linking tables).

- **How to use it:** You must define it at the end of your table creation.

```
CREATE TABLE Enrollments (  
    StudentID INT,  
    CourseID INT,  
    EnrollDate DATE,  
    -- Combining StudentID and CourseID to make one unique Primary Key  
    PRIMARY KEY (StudentID, CourseID)  
);
```

3. ENUM Data Type (The "List" Restriction)

Instead of using a **CHECK** constraint to restrict words (like checking if gender is only 'Male' or 'Female'), you can use the **ENUM** data type. It forces the user to choose a value from a predefined list. If they type anything else, MySQL will show an error.

- **Use cases:** Gender, Status (Active/Inactive), Sizes (Small/Medium/Large).
- **How to use it:**

```
CREATE TABLE Employees (  
    EmpID INT PRIMARY KEY,  
    EmpName VARCHAR(50) NOT NULL,  
    -- The user can ONLY insert 'Male', 'Female', or 'Other'
```

```
Gender ENUM('Male', 'Female', 'Other'),
-- If no status is given, it defaults to 'Active'
Status ENUM('Active', 'Resigned');
);
```

4. Disabling Foreign Key Checks (For Maintenance)

Sometimes in real-world projects, you need to delete (drop) a Parent table or import thousands of rows at once. During this time, Foreign Key constraints will block you and show errors. To fix this, you can **temporarily turn off** the Foreign Key rules, do your work, and then **turn them back on**.

- **Warning:** Use this carefully! Only use it when you are doing system maintenance.
- **How to use it:**

```
-- Step 1: Turn OFF the Foreign Key rules
SET FOREIGN_KEY_CHECKS = 0;

-- Step 2: Do your work here (e.g., Drop tables, insert bulk data)
DROP TABLE Vendors;

-- Step 3: Turn the rules back ON immediately!
SET FOREIGN_KEY_CHECKS = 1;
```